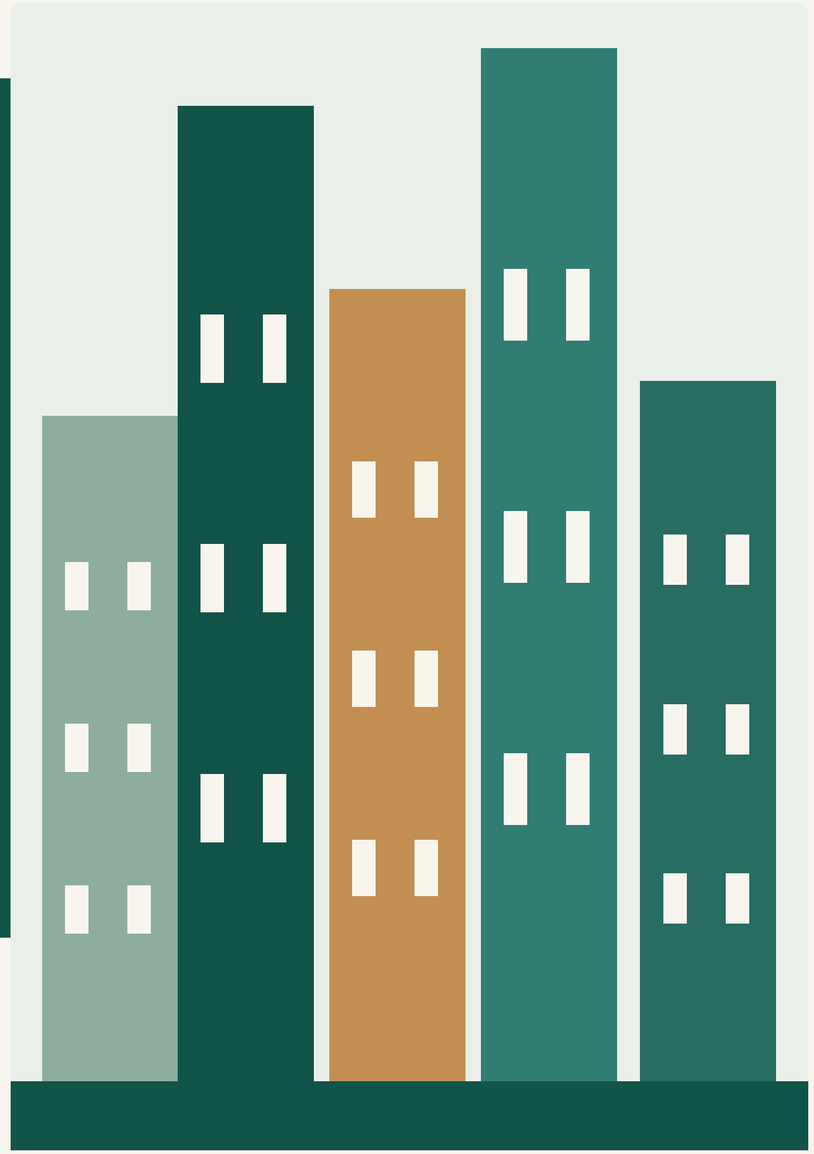
# KAPPA FOUNDATION

Investment recommendation
and revised mandate



**The $12 million allocation that puts our payment dates first.**

# Our recommendation

Meet the obligations without changing the frozen stock weights.

SELECTED POLICY

## Principal matching

**2,400 Beta 1,200 Gamma 2,400 Delta**

**$6.21m**

Net equity investment

**4.80%**

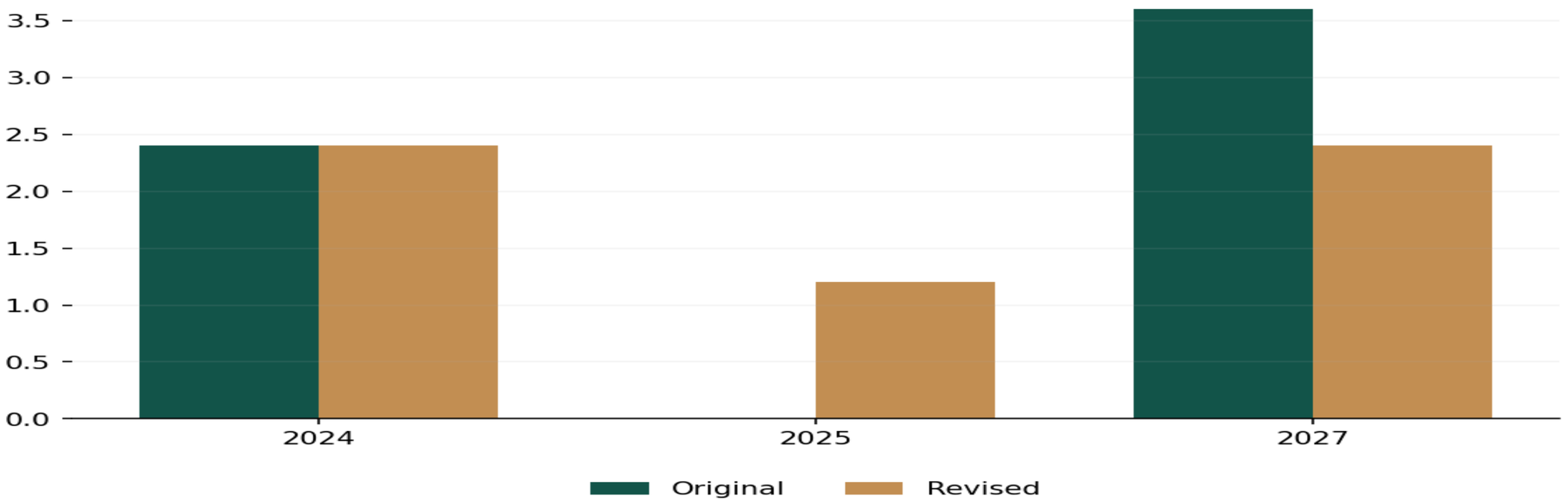
Required equity CAGR to 2030

**A $396k cash reserve remains after the 2025 commitment.**



# The $6m total stays the same

Only the timing of the second commitment has changed.

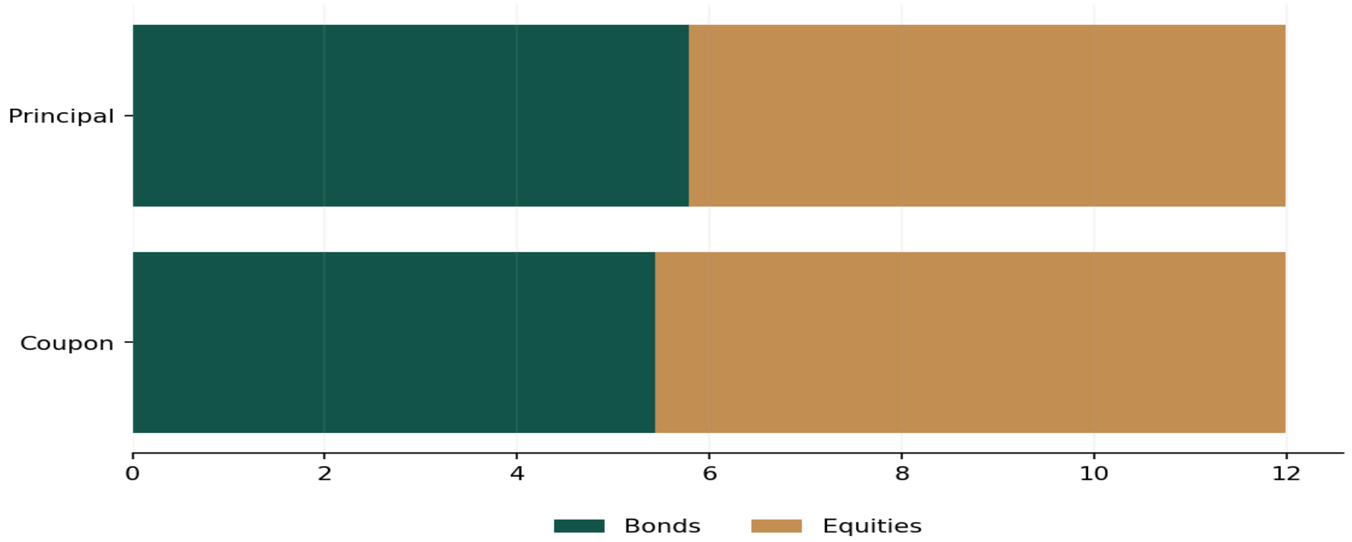



**Revised: $2.4m in 2024, $1.2m in 2025, $2.4m in 2027.**

# We give up growth capital for liquidity

The coupon-funded alternative has more equities but almost no 2025 reserve.

## USD millions committed in 2022




**Coupon funding adds $354k for equities.**

## Cash after the 2025 payment

**$396k**

Principal matching

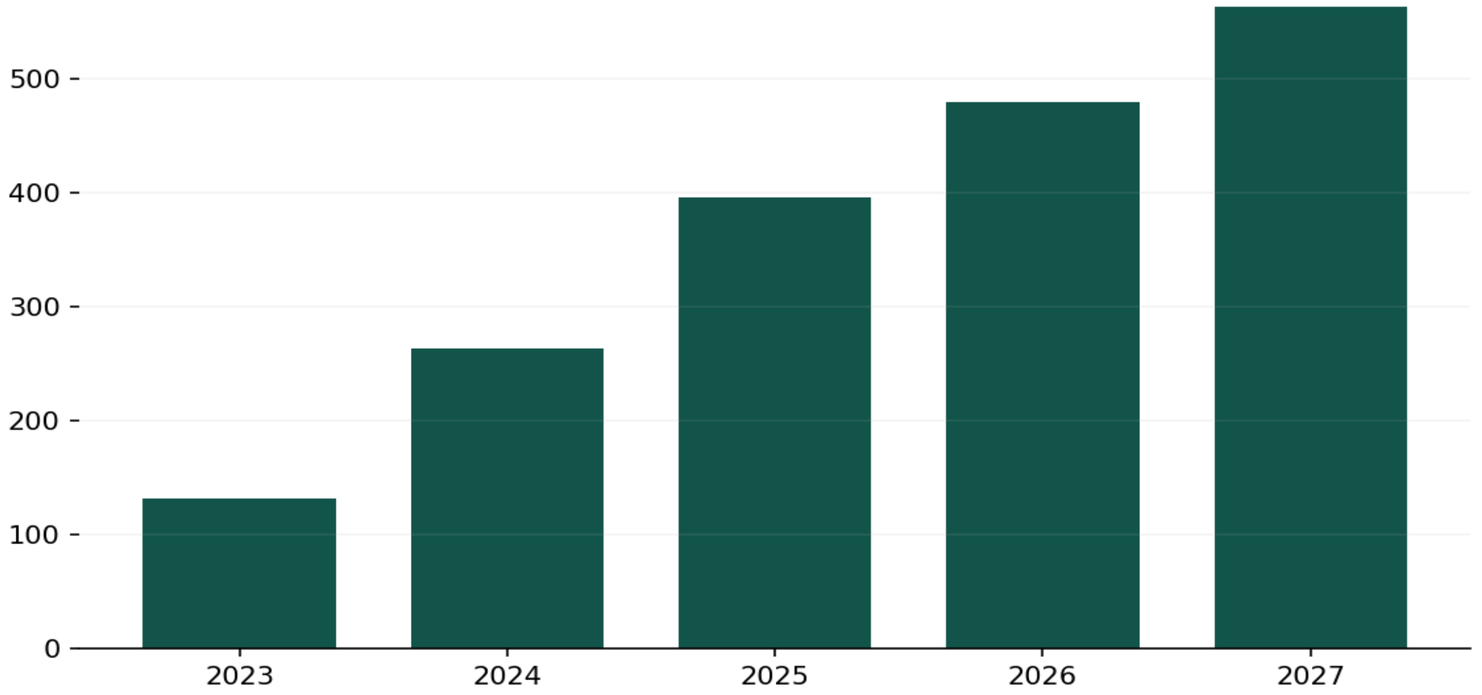
**Nearly zero**

Coupon funding

Both cover the scheduled payments under our cash-flow assumptions.

# The 2025 payment has a cash cushion

Selected bonds mature on time and coupons are retained at zero interest.






## 2025 reserve

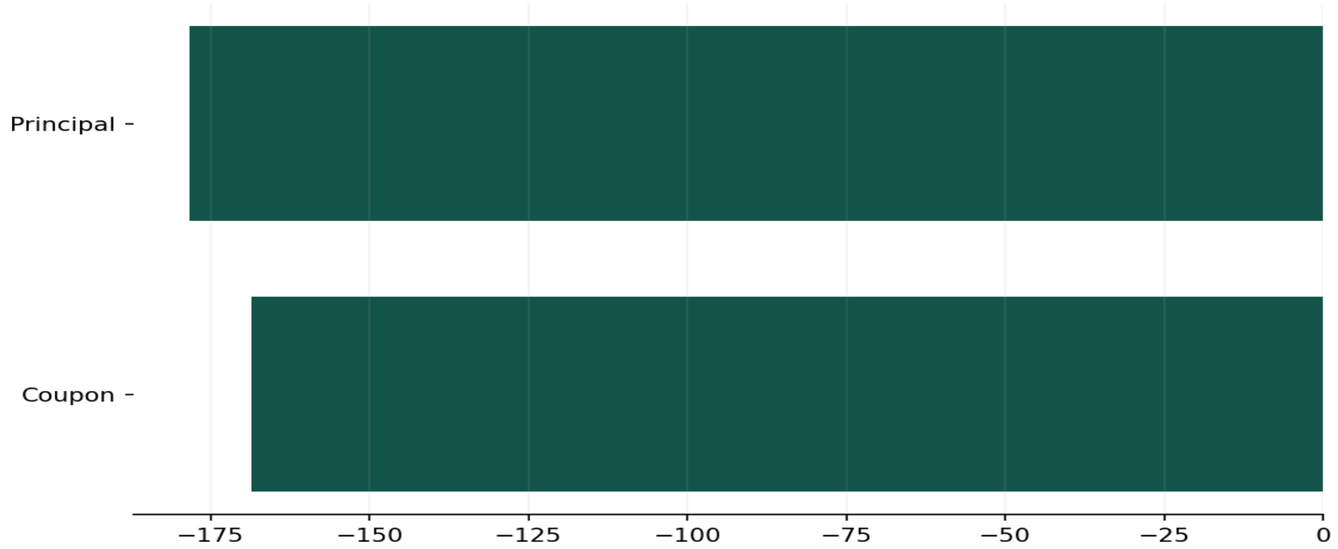
## $396,000

The cash balance is after paying the $1.2m commitment.

# Bond pricing still responds to yields

Holding bonds to maturity reduces funding risk, not market-price volatility.

## Price response to a parallel +100bp shock




**A valuation shock is not a missed bond payment.**

## Case-date present value

$$P = \Sigma\ CFt\ /\ (1 + y)^t$$

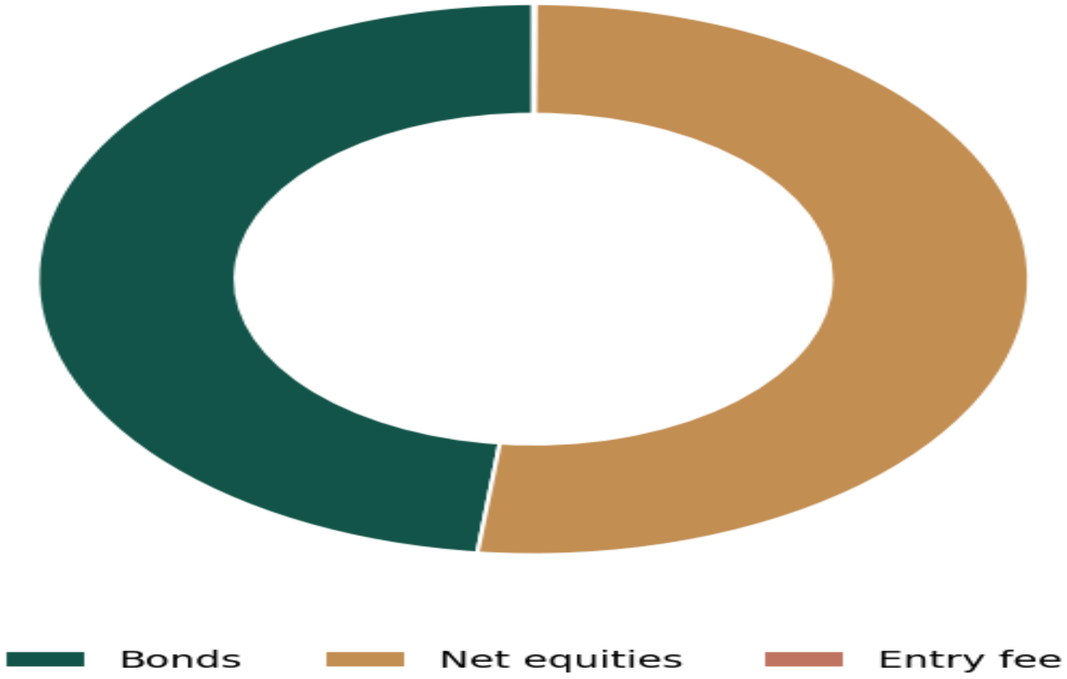
## Rate approximation

$$\Delta P/P \approx -Dmod\ \Delta y + \tfrac{1}{2}\ Convexity\ (\Delta y)^2$$

Bond marks are modeled, not historical market prices.

# Where the full $12 million goes

We show the entry fee separately to keep the allocation accurate.




## Net equity purchased

**$6.210m**

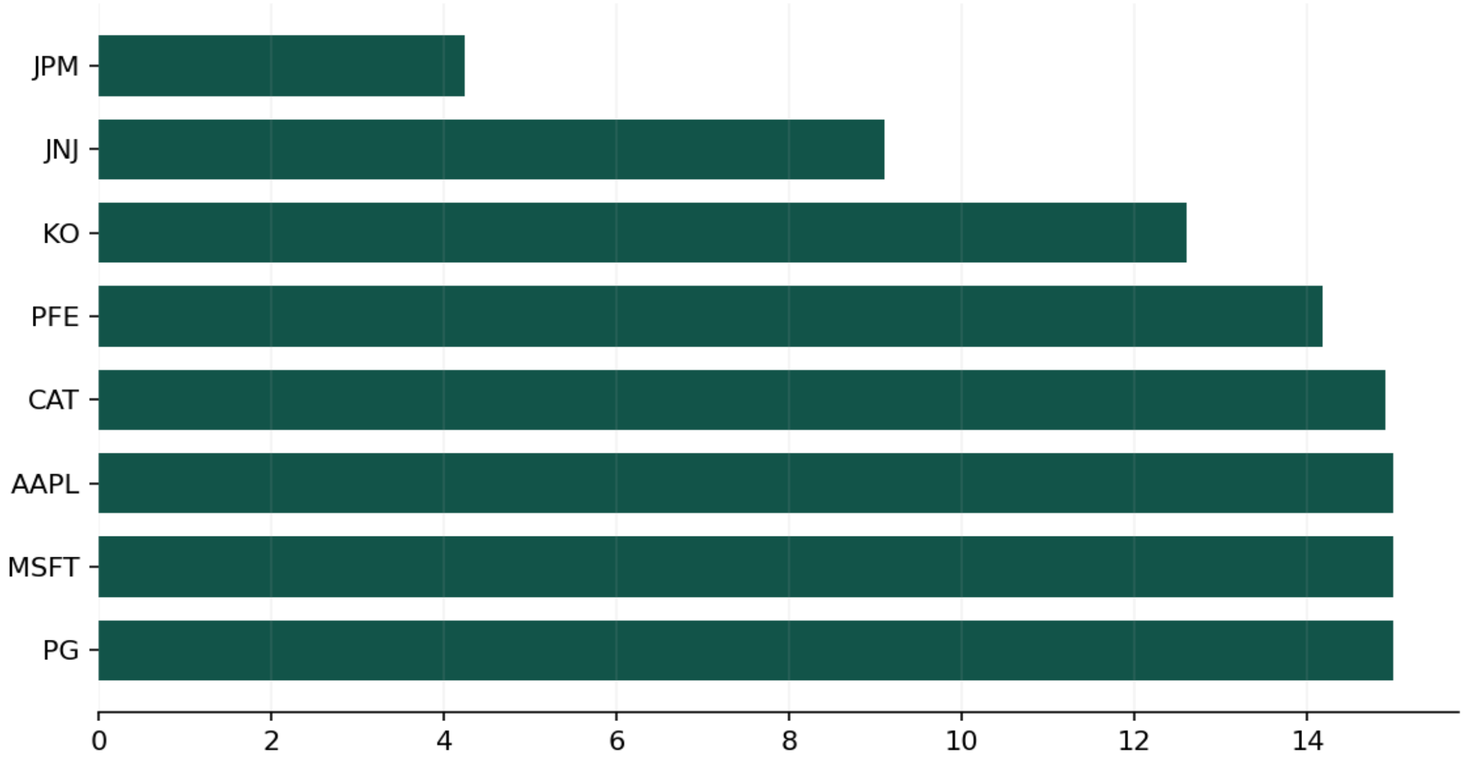
## Entry fee

**E = (12m − Bond Cost) / 1.001**

Ten basis points: about $6,210.

# Eight stocks, with strict weight limits

The candidate list spans six sectors; selected stocks span five.




**Stock cap**

**15%**

**Technology cap**

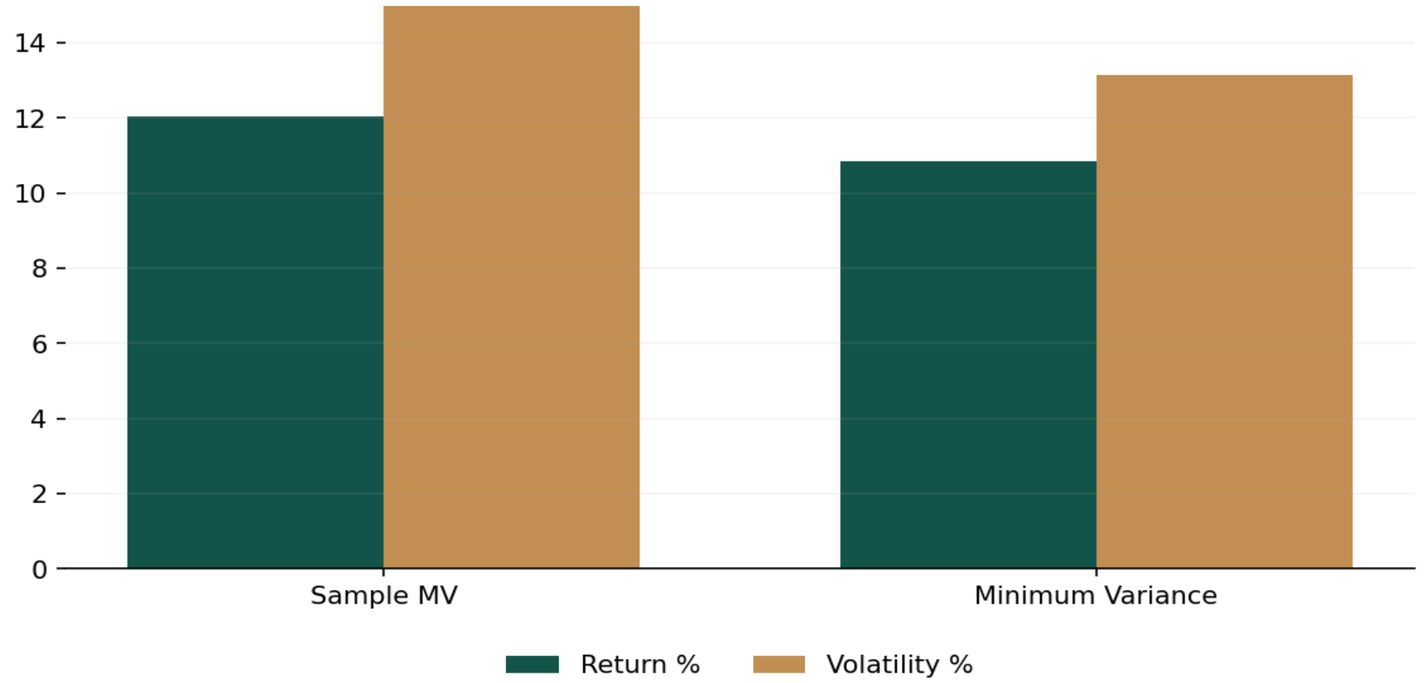
**30%**

**No selected ETFs**

Stock concentration still matters.

# More expected return means more estimated risk

Model statistics are in sample, not observed 2023–2025 returns.




## Sample MV

Full covariance from 96 months

## Single Index

Market-factor covariance assumption

## Minimum Variance

Reduce estimated risk

# What the optimizer actually solves

The forecasts change, but the constraints stay comparable.

## Risk-adjusted objective

$$\max \mu^{T}w - (A/2)\, w^{T}\Sigma w$$

### Investment rules

- $\Sigma w = 1;\ w \geq 0$
- Each direct stock ≤ 15%
- Direct-stock sector ≤ 30%
- Each ETF ≤ 30%; combined ETFs ≤ 40%

## Expected-return blend

$$\mu = 0.25\,\mu_{sample} + 0.75\,\mu_{VTI\ prior}$$

**96 months**

**Jan 2015 – Dec 2022**

Foundation utility also scales by the equity share.

# Re-optimization delivers a tiny modeled gain

Keeping the frozen shares avoids overreacting to unstable forecasts.

## Foundation equity contribution to utility

$$U = F\,\mu p - (A/2)\,F^2\,\sigma p^2$$

A = 3, revised Principal Matching




## Reconsider at higher risk aversion

$A^* \approx 8.94$

Fixed-policy comparison only

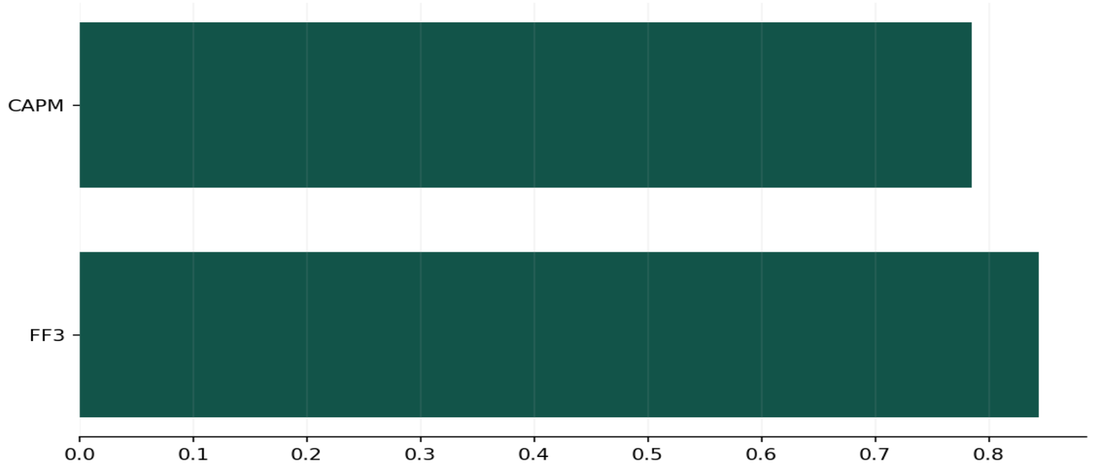
## Reassess and re-optimize

Do not automatically buy the old Minimum Variance portfolio.

# FF3 fits the historical data better

A positive estimated intercept is not proof of future stock-picking skill.

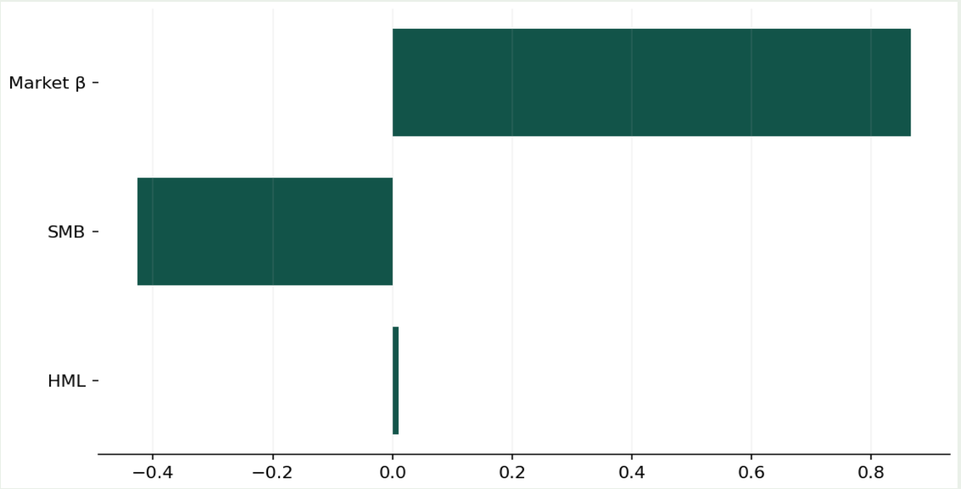
## Adjusted R squared




Excess Rp = α + βm excess Rm + ε

**FF3 adds SMB and HML.**

## FF3 factor loadings

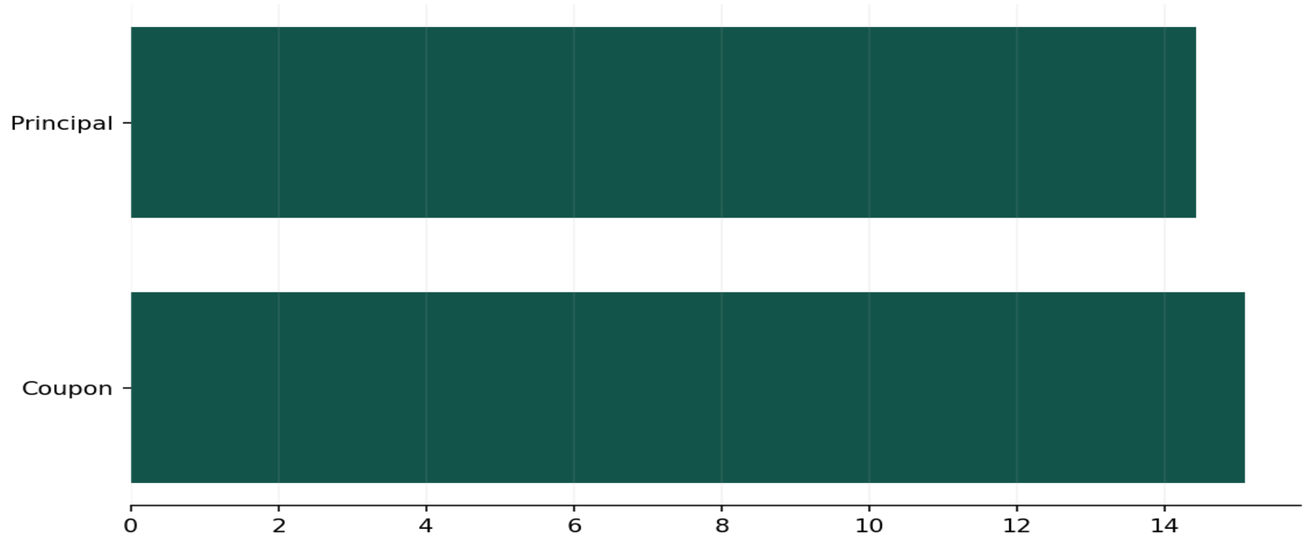



**Monthly FF3 α ≈ +0.569% (in sample)**

# Funding commitments does not eliminate market losses

The 2030 goal also depends on inflation and equity growth.

## Combined shock: stocks −25%, yields +100bp




**A one-time shock is not a rolling 12-month return.**

## 2030 nominal target

**$9.596m**

Assuming 2.3% inflation

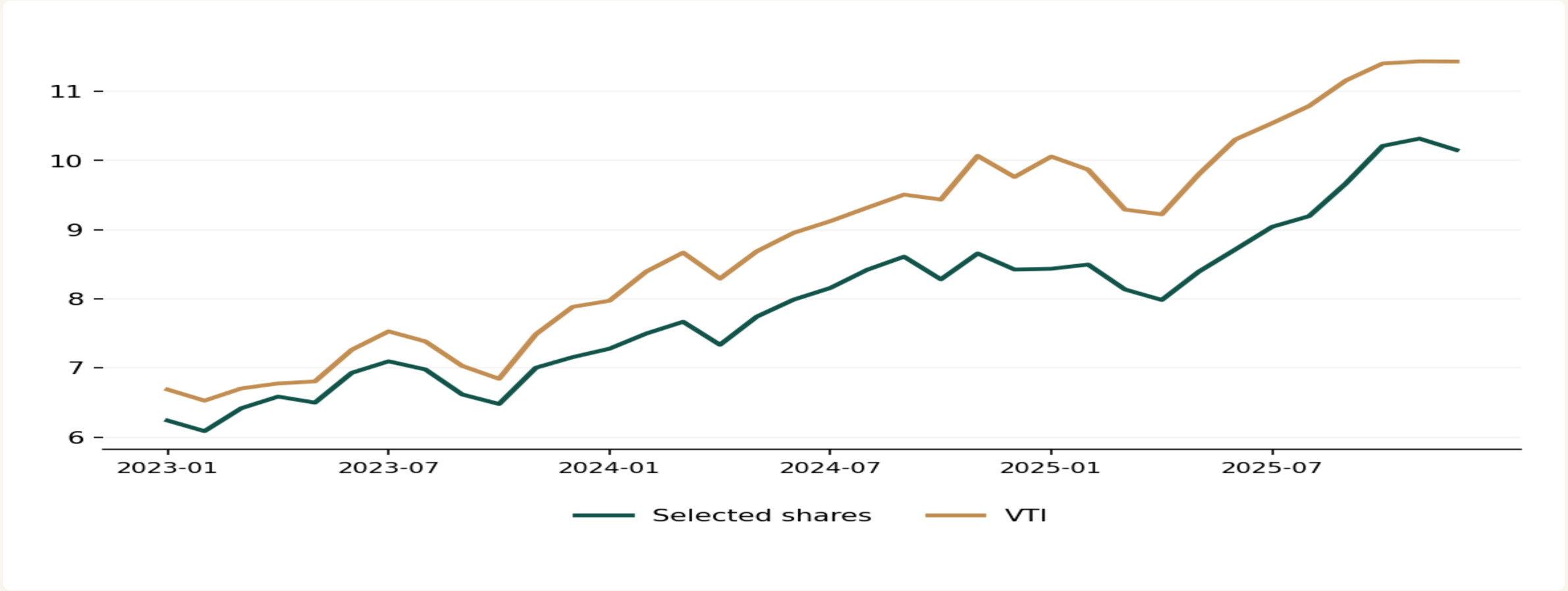
## Equity CAGR needed

**4.80%**

r* = [(Target − Cash) / E]^(1/8) − 1

# The selected shares grew, but VTI did better

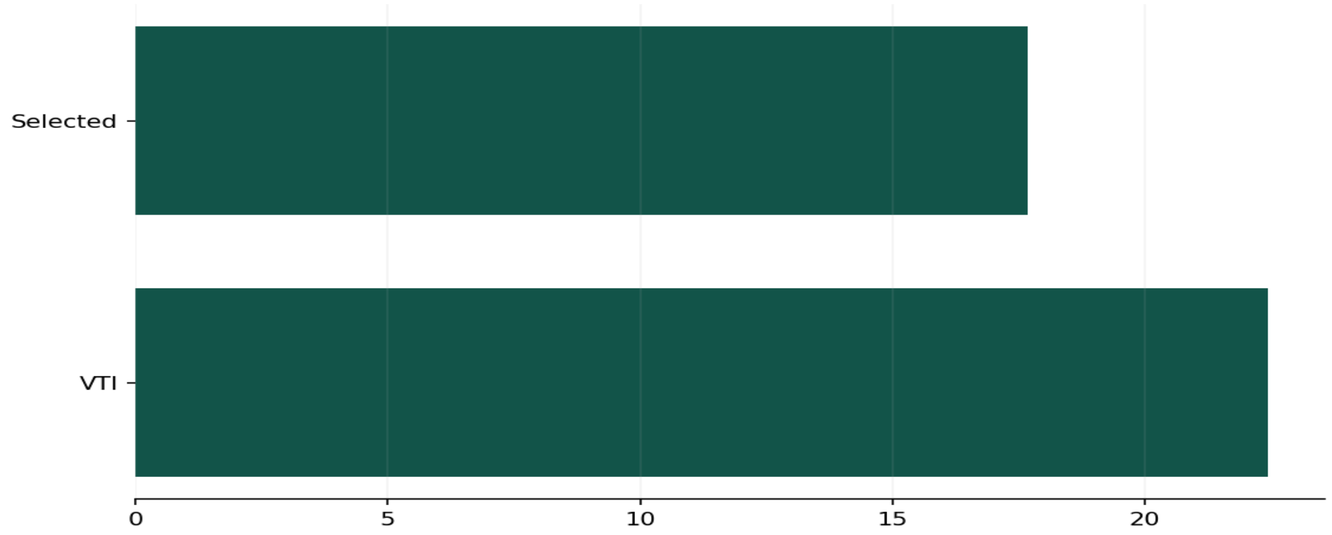
Original $6.229m equity base, not revised $6.210m. Monthly returns, 2023–25.



# Selected shares underperformed the VTI benchmark

Positive absolute returns do not prove an equity-selection advantage.

## CAGR after entry fee




**VTI outperformed by 4.76 percentage points**

## Observed risk

**11.80% vs 12.60%**

Annualized volatility

**6.91%**

Annualized tracking error

**The future test did not change our 2022 decision.**

# Implementation requires monitoring and discipline

Our proposed review rules still require trustee agreement.

| 2024 | 2025 | 2027 |
| --- | --- | --- |
| $2.4m | $1.2m | $2.4m |

## Rolling withdrawal-adjusted review levels

### 12% trailing loss

Warning: review cash and assumptions

### 15% trailing loss

Stronger trustee review

**Rolling TWR = Π(1 + withdrawal-adjusted monthly return) − 1**

# One clear recommendation, with risks disclosed

What we would ask the trustees to approve.



**01**

## Fund the promises

Principal-match every revised payment.

**02**

## Keep the original equity choice

Retain Sample MV weights without hindsight.

**03**

## Monitor and reconsider

Use cash needs and rolling losses to review.

The 2030 goal is uncertain. Human review and university acceptance remain separate evidence.